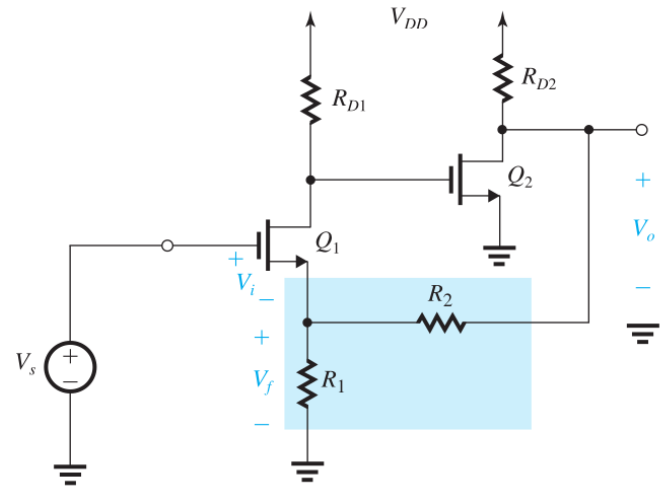


Question 1

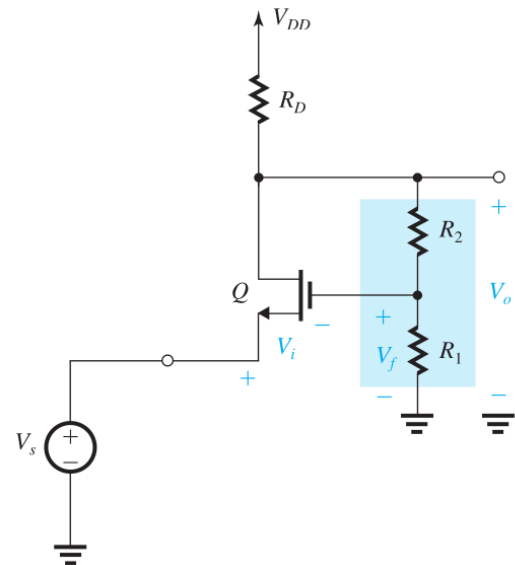
For the series–shunt feedback amplifier of figure below, neglect the MOSFETs' r_o and

- give the feedback network (β circuit) and an expression for β circuit. Also give an expression for the ideal or upper-bound value of the closed-loop gain A_f .
- find the ratio R_2/R_1 that results in an ideal closed-loop gain of 10 V/V. If $R_1 = 1 \text{ k}\Omega$, what value must R_2 have?
- find an expression for the loop gain $A\beta$ circuit.
- if $g_{m1} = g_{m2} = 4 \text{ mA/V}$ and $R_{D1} = R_{D2} = 10 \text{ k}\Omega$, determine the values of $A\beta$ circuit, A , and A_f .

**Question 2**

For the feedback voltage amplifier of figure below:

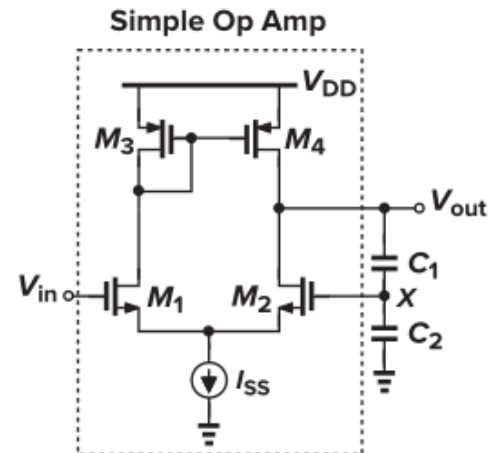
- Find an expression for β .
- Neglecting the MOSFET r_o , find an expression for the loop gain $A\beta$. (Hint: Break the loop at the gate of Q .)
- Find an expression for the open-loop gain A .
- For $g_m = 4 \text{ mA/V}$, $R_D = 10 \text{ k}\Omega$, $R_1 = 20 \text{ k}\Omega$, and $R_2 = 80 \text{ k}\Omega$, find the values of β , $A\beta$, A , and the closed-loop gain A_f . What would A_f be if $A\beta$ were much greater than unity?

**Question 3**

As another example of a series–shunt feedback amplifier, consider the circuit shown in Figure of Question 1, which we analyzed in Question 1 by determining the loop gain $A\beta$. In this example we wish to first analyze the circuit using our systematic procedure and then compare the results to those obtained in Question 1. It is required to obtain the voltage gain V_o/V_s , input resistance R_{in} , and output resistance R_{out} . Find numerical values for the case $g_{m1} = g_{m2} = 4 \text{ mA/V}$, $R_{D1} = R_{D2} = 10 \text{ k}\Omega$, $R_1 = 1 \text{ k}\Omega$, and $R_2 = 9 \text{ k}\Omega$. For simplicity, neglect r_o of each of Q_1 and Q_2 .

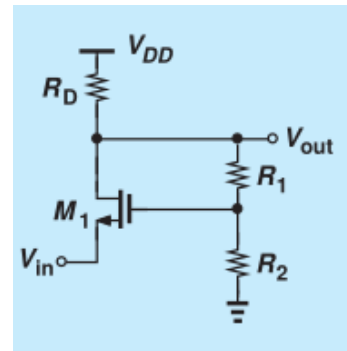
Question 4

Calculate the closed-loop gain and output resistance of the amplifier at relatively low frequencies.



Question 5

Analyze the amplifier depicted in figure below if $R_1 + R_2$ is not much less than R_D .



Question 6

In the given circuit, all three MOSFETs are biased to operate at $g_m = 4 \text{ mA/V}$. The resistance values are $R_{D1} = 10 \text{ kW}$, $R_{D2} = 10 \text{ kW}$, $R_{S1} = 100 \text{ W}$, $R_{S2} = 100 \text{ W}$. Calculate the value of R_F that results in a closed-loop gain is 10 V/V .

